

Notes: Kleven and Waseem (QJE, 2013)
Using Notches to Uncover Optimization Frictions and Structural Elasticities:
Theory and Evidence from Pakistan

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1 Big picture

- **Question.** How can researchers recover structural behavioral elasticities when observed responses are attenuated by optimization frictions such as inattention, adjustment costs, inertia, and misperception?
- **Core empirical object.** The paper studies *notches*: discontinuous jumps in tax liability or benefit rules at a cutoff. In the Pakistani income tax, the average tax rate jumps discretely at bracket cutoffs, so earning one extra rupee can trigger a large discrete increase in tax liability.
- **Why notches are special.** A notch creates both *bunching* below the cutoff and *missing mass* above the cutoff. It also often creates a strictly dominated region above the cutoff, where the taxpayer could lower reported income below the cutoff and get both higher consumption and more leisure.
- **Key idea.** In a frictionless model, nobody should remain in a strictly dominated region. Therefore, observed mass in the dominated region directly reveals optimization frictions. Combining this mass with observed bunching allows the researcher to estimate what bunching would have been absent frictions.
- **Observed response versus structural response.** Observed bunching is an attenuated behavioral response. The structural response is the response that would occur if optimization frictions disappeared. The paper separates these two objects using the bunching-hole structure created by notches.
- **Application.** The authors apply the method to rich administrative income tax data from Pakistan, covering the universe of personal income tax filers from 2006 to 2009.
- **Main result.** Taxpayers bunch sharply below tax notches, but many taxpayers also remain in strictly dominated regions. Frictions are therefore large. Once the authors correct for those frictions, implied frictionless responses are large, but the structural taxable-income elasticities are generally modest.
- **Economic takeaway.** Notches are very inefficient policy instruments. They create extremely high implicit marginal tax rates around cutoffs, so they can induce large behavioral distortions even when underlying structural elasticities are small.

2 Incremental contribution of the paper

- **From kinks to notches.** The earlier bunching literature mostly used kinks, where the marginal tax rate changes discontinuously but tax liability remains continuous. This paper shows that notches are more informative because they generate both bunching and dominated regions.
- **Separating frictions from elasticities.** Kink-based bunching generally confounds small elasticities with large frictions. Kleven and Waseem show that notches can separately identify the observed elasticity, optimization frictions, and the structural elasticity.
- **A nonparametric friction measure.** The paper does not need to specify exactly whether frictions come from adjustment costs, inattention, misperception, or inertia. The density mass

in dominated regions identifies the total amount of friction.

- **A method for structural bounds.** The authors develop two complementary estimators: a lower-bound *bunching-hole method* and an upper-bound *convergence method*. Together they bound the structural elasticity.
- **Developing-country evidence.** The paper provides one of the first applications of modern bunching methods to administrative tax data from a developing country.
- **Evidence on dynamics.** Using repeated observations from 2006 to 2009, the paper shows that dominated behavior declines over time and bunching rises, suggesting that some frictions fade as taxpayers learn or adjust.
- **Policy design contribution.** The paper shows why notched policy schedules can generate large efficiency losses even when conventional elasticities appear small.

3 Data and institutional setting

3.1 Pakistani income tax schedule

- The personal income tax in Pakistan raises a small share of GDP and applies to a selected high-income subset of the population.
- The tax schedule is based on *average tax rates* rather than marginal tax rates. Each bracket has a fixed average tax rate, so moving above a cutoff causes a discrete jump in total tax liability.
- For the self-employed, the average tax rate increases from 0 to 25 percent over 13 notches during 2006–2009.
- For wage earners, the average tax rate increases from 0 to 20 percent over 20 notches, with the analysis focusing mainly on the 2006–2007 schedule.
- At PKR 500,000, earning one more rupee triggers additional tax liability of PKR 12,500 for self-employed individuals and PKR 5,000 for wage earners.
- The tax system also creates an income-composition notch: if self-employment income is at least 50 percent of total income, the taxpayer is classified as self-employed and faces the higher self-employed tax schedule on total income.

Interpretation: The Pakistani tax schedule is an unusually clean environment for studying notches because cutoffs are sharp, the jumps are large, and administrative data reveal the full income distribution around the cutoffs.

3.2 Administrative tax data

- The data come from the Federal Board of Revenue and cover the universe of personal income tax returns filed for tax years 2006–2009.
- The sample contains around 4 million tax-return observations.

- The data include both electronically filed and paper-filed returns. The authors argue that data quality is unusually high for a developing-country tax setting.
- The tax filer population is high income relative to the general population, overwhelmingly male, and selected toward relatively tax-compliant individuals.
- The sample distinguishes wage earners and self-employed individuals based on the share of self-employment income in total income.

Interpretation: The paper estimates behavioral responses among registered tax filers, not among the entire Pakistani labor force. This matters because informal-sector responses and non-filing behavior may be outside the data.

3.3 Enforcement and taxpayer types

- Wage earners are often subject to third-party reporting and withholding by employers. This makes their taxable income harder to manipulate.
- Self-employed individuals face no comparable third-party reporting of taxable income. They are mainly constrained by withholding on certain transactions and by audits, which are infrequent and less effective.
- Self-employed taxpayers are therefore expected to have more flexibility to adjust reported taxable income through evasion, deductions, income timing, or real activity.
- Self-employed taxpayers also exhibit strong round-number reporting, indicating less precise accounting or deliberate reporting at salient values.

Interpretation: The comparison between self-employed and wage earners is informative because it contrasts a flexible, self-reported tax base with a less flexible, third-party-reported tax base.

4 Theory and empirical methodology

4.1 Basic notch model

The model starts from quasi-linear, iso-elastic preferences over consumption and earnings:

$$u = z - T(z) - \frac{n}{1 + 1/e} \left(\frac{z}{n} \right)^{1+1/e},$$

where z is before-tax earnings, $T(z)$ is tax liability, n is ability or potential earnings, and e is the structural earnings elasticity.

Under a proportional tax rate t , the earnings supply condition is

$$z = n(1 - t)^e.$$

This equation defines e as the elasticity of earnings with respect to the marginal net-of-tax rate. Now introduce a notch at earnings cutoff z^* . A general notched tax schedule can be written as

$$T(z) = tz + [\Delta T + \Delta t \cdot z] \mathbf{1}(z > z^*),$$

where ΔT is a pure tax-liability jump and Δt is a proportional tax-rate jump above the cutoff.

Interpretation: The notch does not merely change the slope of the budget set. It creates a discrete drop in after-tax income at the cutoff. This makes the local implicit marginal tax rate extremely large, often above 100 percent.

4.2 Dominated range and bunching range

A downward tax notch creates two key regions:

- **Dominated range:** an interval $(z^*, z^* + \Delta z_D]$ above the cutoff where the taxpayer can move to z^* and obtain both higher consumption and more leisure.
- **Bunching range:** an interval $(z^*, z^* + \Delta z^*]$ of pre-notch incomes whose agents choose to locate at the cutoff after the notch is introduced.

The bunching range is at least as wide as the dominated range:

$$\Delta z^* \geq \Delta z_D.$$

Even with a zero compensated elasticity, a notch generates bunching from the dominated region. This is fundamentally different from a kink, where zero elasticity implies zero bunching.

Interpretation: The dominated range is the paper's key empirical innovation. In a frictionless world, this region must be empty. If it is not empty in the data, the remaining mass reveals optimization frictions.

4.3 Structural elasticity from the marginal buncher

The marginal buncher is the individual who is just indifferent between bunching at z^* and choosing the best interior point above the notch. This indifference condition yields an implicit equation linking the earnings response $\Delta z^*/z^*$, the size of the notch, and the structural elasticity e :

$$\frac{1}{1 + \Delta z^*/z^*} \left(1 + \frac{\Delta T/z^*}{1 - t} \right) - \frac{1}{1 + 1/e} \left(\frac{1}{1 + \Delta z^*/z^*} \right)^{1+1/e} - \frac{1}{1 + e} \left(\frac{1 - t - \Delta t}{1 - t} \right)^{1+e} = 0.$$

Given the tax parameters and an estimate of Δz^* , the structural elasticity e can be solved numerically.

Interpretation: The estimation problem is transformed into estimating the response of the marginal buncher. Once that response is known, the elasticity follows from the geometry of the notched budget set.

4.4 Bunching and the counterfactual density

Let $h_0(z)$ be the counterfactual earnings density in the absence of the notch. Excess bunching is

$$B = \int_{z^*}^{z^* + \Delta z^*} h_0(z) dz \approx h_0(z^*) \Delta z^*.$$

Thus, observed bunching gives an estimate of the earnings response if the counterfactual density is smooth around the cutoff.

The counterfactual distribution is estimated by fitting a flexible polynomial to the empirical density while excluding the region affected by bunching and missing mass:

$$c_j = \sum_{i=0}^p \beta_i z_j^i + \sum_{i=z_L}^{z_U} \gamma_i \mathbf{1}(z_j = i) + \varepsilon_j.$$

The fitted counterfactual is obtained from the polynomial part, omitting the excluded-range dummies.

Interpretation: The empirical challenge is to estimate what the density would have looked like without the notch. The paper uses nearby density information while excluding the notch-affected region.

4.5 Optimization frictions

Let a^* denote the share of individuals in the dominated region who remain unresponsive. This is estimated by

$$a^* = \frac{\text{observed mass in dominated region}}{\text{counterfactual mass in dominated region}}.$$

Observed bunching equals structural bunching attenuated by the share of responders:

$$B \approx h_0(z^*)(1 - a^*)E[\Delta z_e^*].$$

Therefore, structural bunching absent frictions is estimated by inflating observed bunching:

$$B^{structural} = \frac{B}{1 - a^*}.$$

Interpretation: If 70 percent of taxpayers in a dominated region are unresponsive, then observed bunching captures only about 30 percent of the frictionless response. This is why the paper's correction for frictions is quantitatively important.

4.6 Lower and upper bounds

The paper uses two methods:

- **Bunching-hole method:** scales observed bunching by $1/(1 - a^*)$. This provides a lower bound on the average structural response under plausible assumptions.
- **Convergence method:** uses the point z_U where observed and counterfactual densities converge. This provides an upper bound on the structural response, especially when elasticities are heterogeneous.

Interpretation: The bunching-hole method uses very local information around the notch, whereas the convergence method uses the broader missing-mass region. The two estimates together bound the structural elasticity.

4.7 Round-number reporting

Self-employed taxpayers report taxable income at round numbers. Because tax notches are also located at round numbers, raw excess mass at notches would overstate true notch bunching.

The authors address this using round-number fixed effects:

$$c_j = \sum_{i=0}^p \beta_i z_j^i + \sum_{r \in R, 12R} \delta_r \mathbf{1}\left(\frac{z_j}{r} \in \mathbb{N}\right) + \sum_{i=z_L}^{z_U} \gamma_i \mathbf{1}(z_j = i) + \varepsilon_j,$$

where $R = \{1K, 5K, 10K, 25K, 50K\}$ captures annual and monthly rounding patterns.

Interpretation: This adjustment is crucial for the full self-employed sample. Otherwise, the method would confuse tax-driven bunching with generic reporting at salient numbers.

5 Identification and empirical design

5.1 What the design identifies

The empirical design identifies three related objects:

- **Observed elasticity:** the behavioral response seen in the data, attenuated by optimization frictions.
- **Optimization frictions:** the share of taxpayers in dominated regions who remain unresponsive.
- **Structural elasticity:** the elasticity that would govern behavior absent optimization frictions, interpreted as a long-run or frictionless response parameter.

5.2 Key assumptions

The paper’s notch design relies on three main assumptions:

1. **Smooth counterfactual density.** In the absence of the notch, the income distribution would be smooth around the cutoff.
2. **Continuous set of bunchers.** Bunchers come from a continuous interval above the cutoff.
3. **Locally constant frictions.** The share of unresponsive individuals estimated in the dominated region is informative about frictions over the nearby bunching range.

Interpretation: These assumptions are still weaker than the assumptions required in kink-based bunching designs, because the notch provides an independent empirical moment - mass in the dominated region - to measure frictions.

5.3 Limitations of the design

- The method is designed to estimate intensive-margin taxable-income responses near the notch.

- It does not fully capture extensive responses such as formal-informal switching or the decision to file a tax return.
- It may not capture taxpayers who overshoot the notch and move far below the cutoff because of coarse reporting or adjustment frictions.
- The results apply to the tax-filing population, not necessarily to all workers or firms in Pakistan.

Interpretation: The method is powerful for identifying local optimization frictions and structural intensive elasticities, but it is not a complete model of all tax responses in a developing economy.

6 Tables and figures

6.1 Figure I: Behavioral Responses to a Tax Notch

Read:

- Panel A shows how a tax notch creates a discrete drop in the budget set at earnings z^* .
- The individual just above the cutoff can often increase both consumption and leisure by moving down to the cutoff.
- Panel B maps the budget-set geometry into the density distribution: bunching below the cutoff and a hole above it.
- The dominated range is strictly smaller than the full bunching range.

Economic message: A notch is much more than a large kink. It creates a dominated region. This dominated region makes optimization frictions visible in the data.

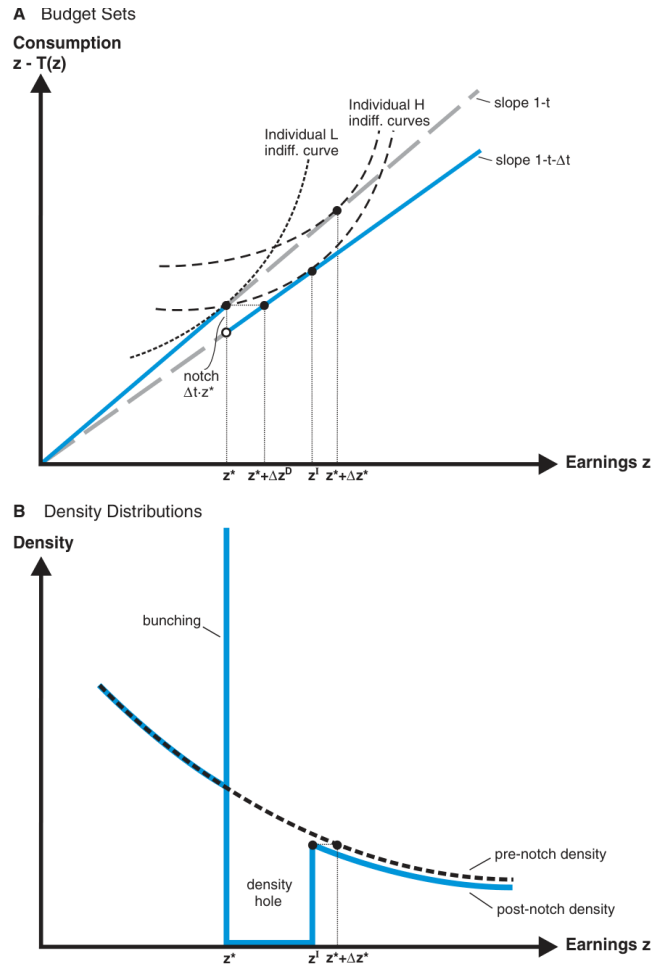


FIGURE I
Behavioral Responses to a Tax Notch

Figure I: Behavioral Responses to a Tax Notch.

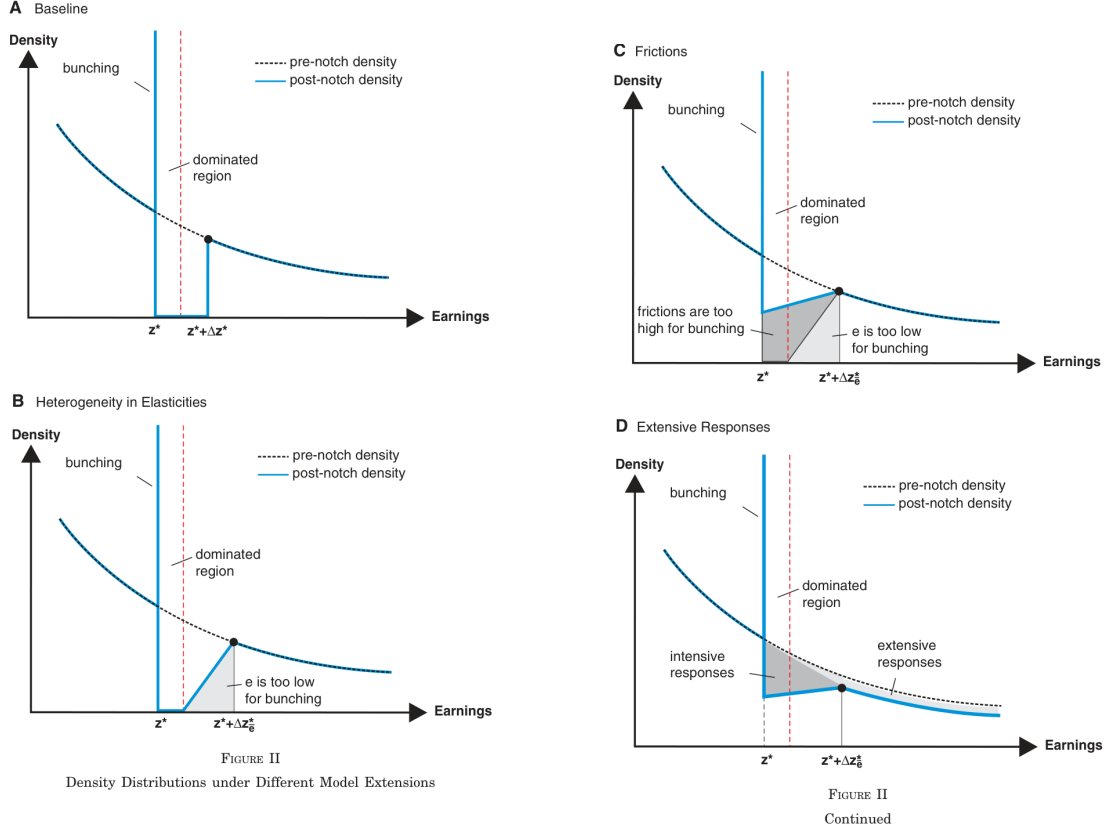
6.2 Figure II: Density Distributions under Different Model Extensions

Read:

- Panel A gives the baseline density response: sharp bunching at the cutoff and missing mass above it.
- Panel B introduces heterogeneous elasticities. Low-elasticity individuals do not bunch even in a frictionless model, so the post-notch density rises gradually above the notch.
- Panel C introduces optimization frictions. Some individuals who should respond remain in dominated or affected regions.
- Panel D introduces extensive responses, which affect the distribution farther away from the cutoff but not the local dominated range in the same way.

Economic message: The shape of the density around a notch contains information about the

anatomy of behavior: true elasticities, frictions, and extensive responses leave different empirical footprints.



(a) Figure II, Panels A and B: Baseline and heterogeneous elasticities.

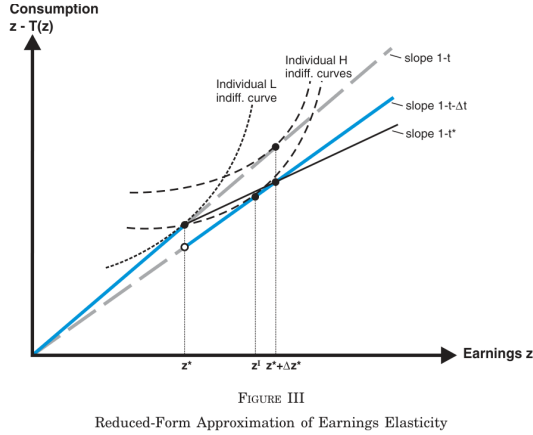
(b) Figure II, Panels C and D: Frictions and extensive responses.

6.3 Figure III and Figure IV: Reduced-Form Elasticity and Counterfactual Estimation

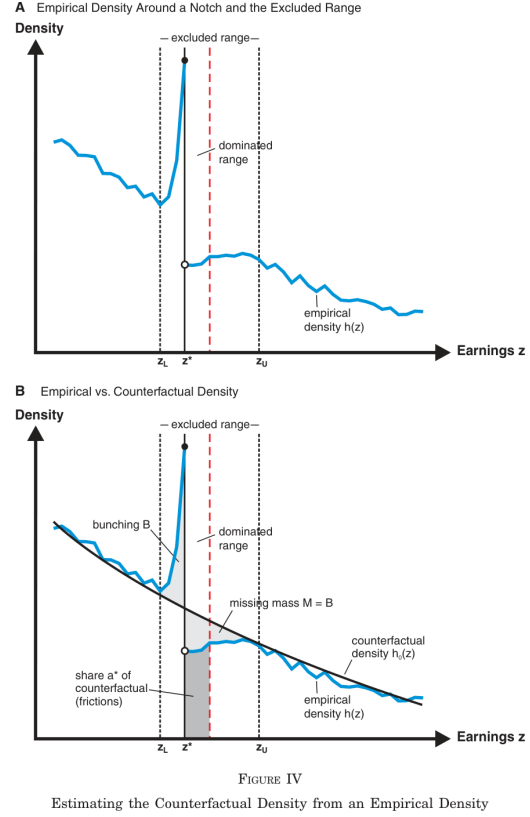
Read:

- Figure III shows the reduced-form method, which treats the notch as if it were a hypothetical kink with an implicit marginal tax rate.
- The reduced-form elasticity is an approximation and generally an upper bound.
- Figure IV shows how the empirical density is compared with a smooth counterfactual density.
- The upper bound z_U of the excluded range is chosen so that missing mass equals excess bunching.

Economic message: The paper is careful about translating bunching into elasticities: it does not simply count mass at the cutoff, but reconstructs the counterfactual density and uses the missing-mass area to discipline the response range.



(a) Figure III: Reduced-form approximation of earnings elasticity.



(b) Figure IV: Estimating the counterfactual density.

6.4 Figure V: Personal Income Tax Schedules in Pakistan

Read:

- The solid line shows the self-employed schedule, with much higher average tax rates and large jumps at bracket cutoffs.
- The dashed line shows the wage-earner schedule, which also contains notches but with smaller tax jumps.
- The cutoff itself belongs to the low-tax side of the notch for these income-level notches.
- The self-employed schedule creates stronger incentives for bunching and missing mass.

Economic message: The Pakistani system creates unusually strong quasi-experimental variation. The tax liability jumps are large enough that taxpayers should respond visibly unless they face substantial frictions.

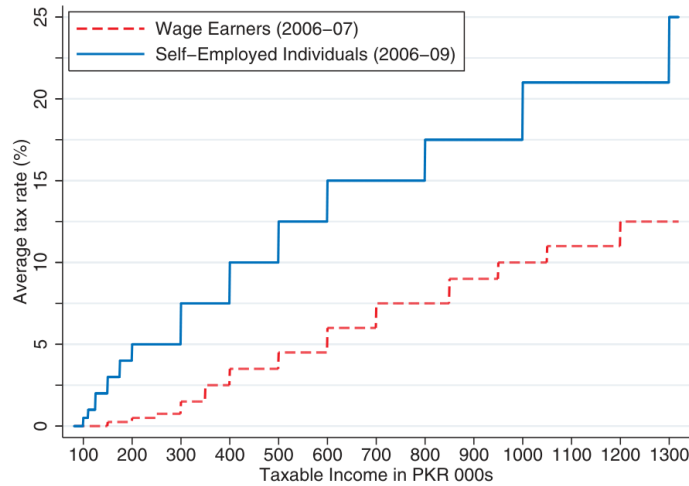


FIGURE V
Personal Income Tax Schedules in Pakistan

The figure shows the average tax rate as a function of annual taxable income for wage earners in 2006–7 (dashed line) and for self-employed individuals in 2006–9 (solid line). Taxable income is shown in thousands of Pakistani rupees (PKR), with the PKR-USD exchange rate varying from 60 to 80 during these years. Each bracket cutoff is associated with a discrete jump in the average tax rate (a notch), and the cutoff itself belongs to the lowtax side of the notch. The tax rate on self-employed individuals increases from 0 to 25% over

Figure V: Personal Income Tax Schedules in Pakistan.

6.5 Figure VI: Self-Employed Individuals, Non-Rounder Sample

Read:

- Panels A and B show the empirical distribution around the first ten notches for self-employed non-rounders.
- There is sharp excess mass just below each notch and missing mass just above each notch.
- Panels C–F compare empirical and counterfactual densities at the 300K, 400K, 500K, and 600K notches.
- The estimates show large bunching, but also large shares of unresponsive taxpayers in dominated ranges.

Economic message: Self-employed taxpayers clearly respond to tax notches, but the response is heavily attenuated. The fact that many taxpayers remain in dominated regions means frictions are first-order.

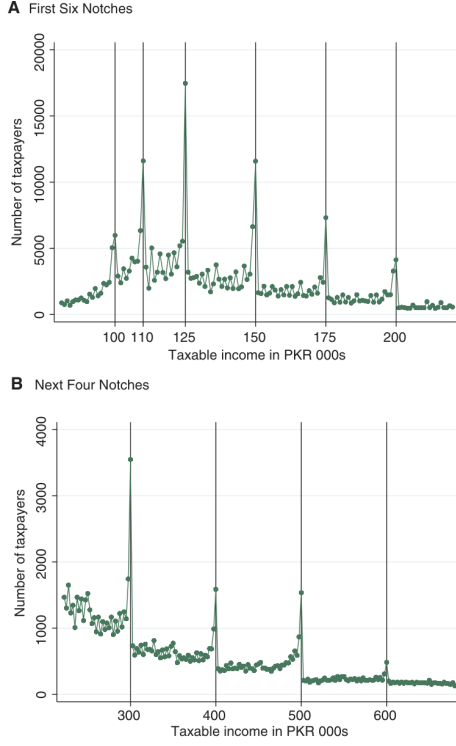


FIGURE VI

Empirical and Counterfactual Distributions around Notches: Self-Employed

(a) Figure VI, Panels A and B: Broad self-employed non-rounder distribution.

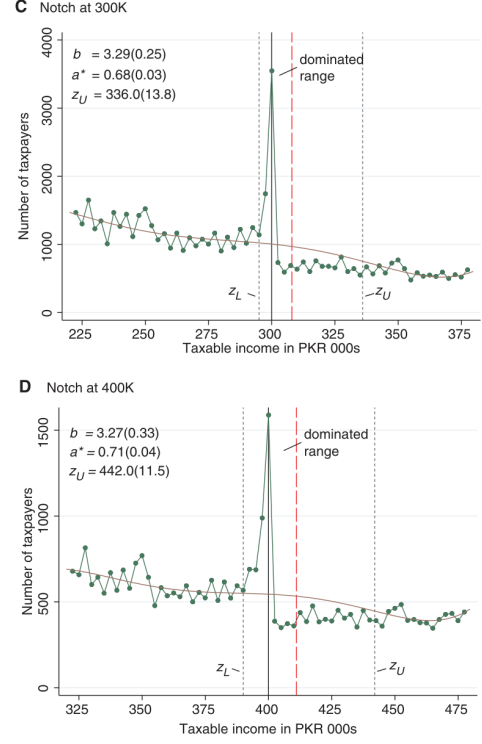


FIGURE VI

Continued

dominated regions are marked by vertical long-dashed lines, and excluded ranges $[z_L, z_U]$ are marked by vertical short-dashed lines. Bunching b is excess mass in the excluded range below the notch (in proportion to the average counterfactual frequency in the dominated range), a^* is the share of individuals

(b) Figure VI, Panels C and D: Notches at 300K and 400K.

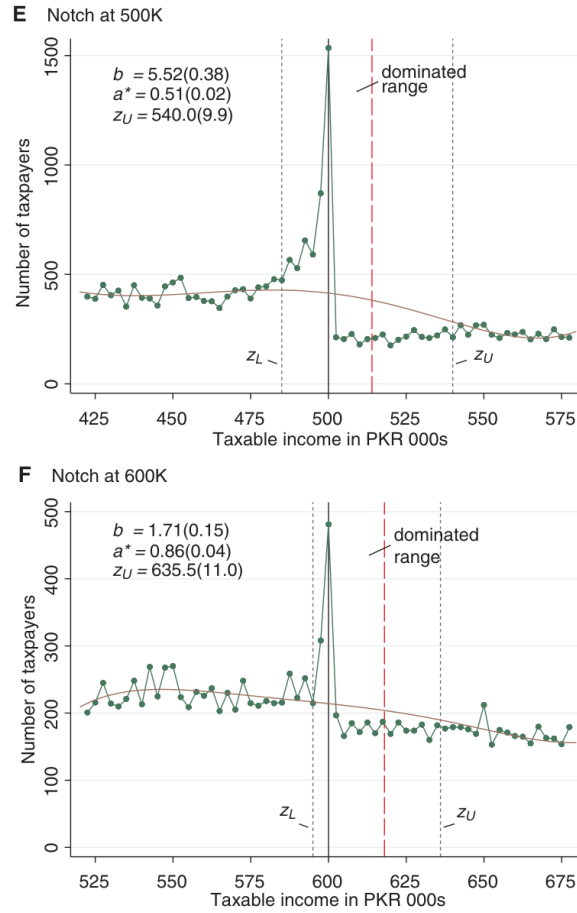


Figure VI, Panels E and F: Notches at 500K and 600K.

6.6 Figure VII: Self-Employed Individuals, Full Sample

Read:

- The full sample includes both rounders and non-rounders.
- The distribution contains excess mass at notches and at other round numbers.
- The counterfactual controls for round-number bunching using annual and monthly round-number fixed effects.
- The qualitative findings remain similar to the non-rounder sample: bunching is large and frictions remain large.

Economic message: The tax-response evidence is not an artifact of excluding rounders. However, the full-sample estimates require stronger controls because self-employed taxpayers report income at salient numbers even away from notches.

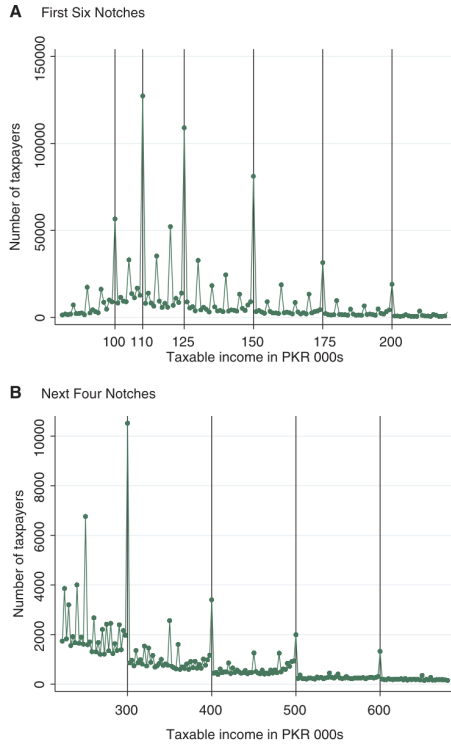


FIGURE VII

Empirical and Counterfactual Distributions around Notches: Self-Employed Individuals (Full Sample)

(a) Figure VII, Panels A and B: Full-sample self-employed distribution.

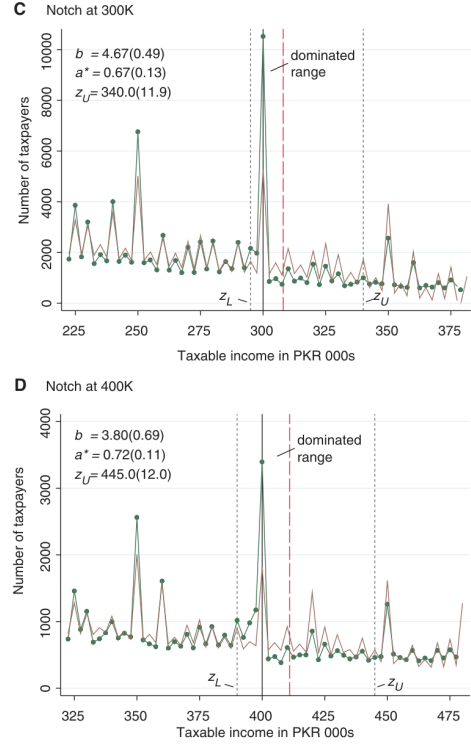


FIGURE VII

Continued

bounds of dominated regions are marked by vertical long-dashed lines, and excluded ranges $[z_L, z_U]$ are marked by vertical short-dashed lines. Bunching k is excess mass in the excluded range below the notch (in proportion to the

(b) Figure VII, Panels C and D: Full-sample notches at 300K and 400K.

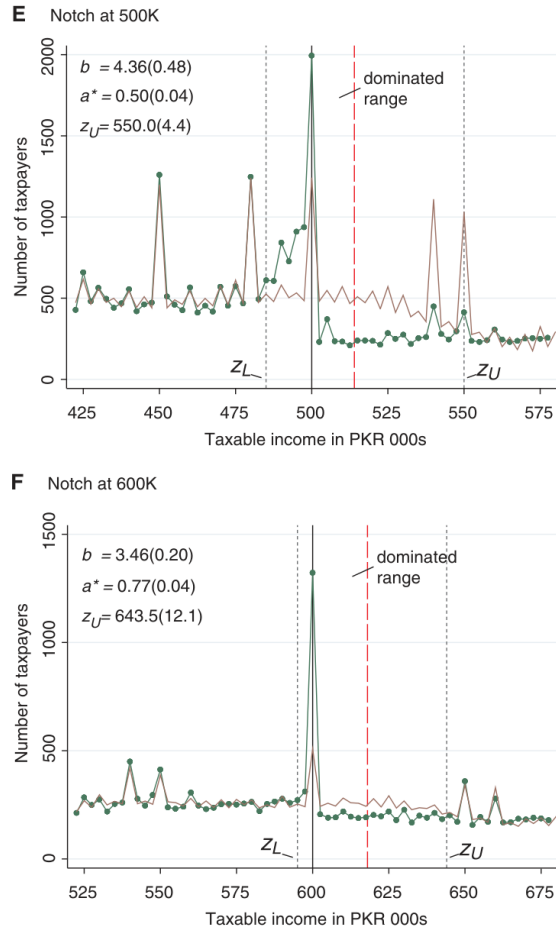


Figure VII, Panels E and F: Full-sample notches at 500K and 600K.

6.7 Table I: Dynamics of Dominated, Bunching, and Inconsistent Behavior

Read:

- Table I tracks dominated behavior, bunching behavior, and inconsistent reporting from 2006 to 2009.
- Dominated behavior declines over time, especially in the balanced panel of repeated filers.
- Bunching behavior rises over time and is more persistent than dominated behavior.
- Inconsistent reporting is much more common among taxpayers in dominated regions than among bunchers.

Economic message: Frictions are not purely permanent. They decline with experience, filing history, and learning. This supports the interpretation of structural elasticities as closer to long-run responses than observed short-run elasticities.

Year (1)	#Obs. (2)	Dominated Behavior				Bunching Behavior				Inconsistent Reporting		
		(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12) Among Dominated	(13) Among Bunchers
		Total	2-Year	3-Year	4-Year	Total	2-Year	3-Year	4-Year	Total		
Panel A: Non-rounder sample												
A1: Unbalanced panel												
2006	136,037	5.1	-	-	-	35.3	-	-	-	12.3	17.1	9.4
		(0.96)				(0.12)				(0.09)	(0.46)	(0.17)
2007	129,443	4.4	0.43	-	-	35.2	6.1	-	-	10.6	15.0	8.1
		(0.96)	(0.018)			(0.12)	(0.07)			(0.09)	(0.47)	(0.15)
2008	130,110	4.4	0.36	0.09	-	35.9	7.3	2.2	-	10.2	15.1	8.6
		(0.96)	(0.017)	(0.008)		(0.12)	(0.07)	(0.04)		(0.08)	(0.44)	(0.15)
2009	108,321	4.6	0.20	0.06	0.02	27.7	6.9	2.5	0.9	15.3	20.1	5.1
		(0.96)	(0.017)	(0.007)	(0.005)	(0.14)	(0.08)	(0.05)	(0.03)	(0.11)	(0.65)	(0.13)
A2: Balanced panel												
2006	30,120	4.3	-	-	-	35.2	-	-	-	11.6	14.7	9.3
		(0.12)				(0.25)				(0.18)	(0.39)	(0.33)
2007	30,120	5.6	0.47	-	-	27.2	10.7	-	-	11.7	16.1	9.0
		(0.11)	(0.039)			(0.26)	(0.18)			(0.19)	(1.11)	(0.32)
2008	30,120	5.8	0.37	0.18	-	27.4	11.9	5.2	-	11.7	16.9	9.6
		(0.11)	(0.035)	(0.024)		(0.26)	(0.19)	(0.13)		(0.19)	(1.11)	(0.32)
2009	30,120	5.6	0.41	0.12	0.11	26.9	13.5	6.5	3.1	7.1	16.1	5.0
		(0.11)	(0.037)	(0.020)	(0.019)	(0.27)	(0.20)	(0.14)	(0.10)	(0.15)	(1.12)	(0.17)

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Year (1)	#Obs. (2)	Dominated Behavior				Bunching Behavior				Inconsistent Reporting		
		(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
		Total	2-Year	3-Year	4-Year	Total	2-Year	3-Year	4-Year	Total	Dominated	Bunchers
Panel B: Full sample												
B1: Unbalanced panel												
2006	405,200	2.5	-	-	-	33.5	-	-	-	8.4	15.1	7.0
		(0.02)				(0.07)				(0.04)	(0.36)	(0.07)
2007	393,925	2.1	0.22	-	-	35.2	13.6	-	-	7.8	14.1	6.3
		(0.02)	(0.007)			(0.08)	(0.05)			(0.04)	(0.38)	(0.06)
2008	379,962	2.2	0.18	0.04	-	36.0	15.6	7.1	-	8.2	12.9	6.4
		(0.02)	(0.007)	(0.003)		(0.08)	(0.06)	(0.04)		(0.04)	(0.37)	(0.07)
2009	324,901	2.2	0.15	0.02	0.01	40.7	15.4	7.9	4.0	8.5	26.9	3.3
		(0.03)	(0.007)	(0.003)	(0.002)	(0.09)	(0.06)	(0.05)	(0.03)	(0.05)	(0.51)	(0.05)
B2: Balanced panel												
2006	167,500	2.5	-	-	-	35.0	-	-	-	8.2	13.2	6.9
		(0.04)				(0.12)				(0.07)	(0.32)	(0.10)
2007	167,500	1.9	0.26	-	-	36.7	17.7	-	-	8.0	13.8	6.5
		(0.05)	(0.012)			(0.12)	(0.09)			(0.07)	(0.61)	(0.10)
2008	167,500	1.9	0.18	0.06	-	37.3	18.4	10.8	-	8.7	14.4	6.8
		(0.03)	(0.010)	(0.006)		(0.12)	(0.10)	(0.08)		(0.07)	(0.62)	(0.10)
2009	167,500	1.8	0.19	0.04	0.02	43.9	21.7	12.8	7.7	9.7	11.4	1.9
		(0.03)	(0.011)	(0.005)	(0.004)	(0.12)	(0.10)	(0.08)	(0.06)	(0.05)	(0.57)	(0.05)

Notes: The table shows the dynamics of dominated, bunching, and inconsistent behavior for self-employed individuals from 2006 to 2009. Panel A and B show the non-rounder and full samples, respectively, and each panel distinguishes between the unbalanced panel (everybody filing at least once in the sample period) and the balanced panel (those filing every year in the sample period). The table shows the total fraction featuring dominated or bunching behavior in each year as well as the fraction featuring such behavior for two, three, or four consecutive years. Bunchers include everybody listing in the bunching range below one of the 13 notches, only a subset of whom are cross bunchers actively responding to the tax system. The table overstates misreporting as a determinant of dominated behavior, since misreporters' between self-assessed tax liability and health income

NOTES AND OPTIMIZATION FRICTIONS

(a) Table I, part 1: Dynamics for self-employed tax-payers.

(b) Table I, part 2: Continued dynamics and notes.

6.8 Table II and Table III: Structural Elasticities for Self-Employed Individuals and Wage Earners

Read:

- Table II estimates self-employed structural elasticities using the non-rounder sample.
- For self-employed individuals, friction shares in dominated regions are roughly 50–86 percent depending on the notch.
- Corrected earnings responses are large, usually around 5–15 percent of total earnings.
- Structural elasticities are generally modest: lower-bound estimates mostly fall around 0.05–0.15, with a larger estimate at the 200K notch.
- Table III repeats the exercise for wage earners. Frictions are even larger - around 87–91 percent.
- Wage-earner structural elasticities are very small, generally around 0.05 or lower.

Economic message: Large bunching does not imply large structural elasticities. In this setting, large responses arise because notches create extreme incentives, while underlying elasticities are modest and observed responses are sharply attenuated by frictions.

TABLE II STRUCTURAL EARNINGS ELASTICITIES FOR SELF-EMPLOYED INDIVIDUALS (NON-ROUNDER SAMPLE)										
(1) Notch	(2) ATR Point	(3) Dominated Range Δx	Frictions α^a		Earnings Response Δx^a		Structural Elasticity ϵ		Reduced-Form Elasticity ϵ_R	
			(4) Full Range Δx^b	(5) Lower Range $\Delta x^{b/2}$	(6) Bunching-Hole Method	(7) Conv. Method	(8) Bunching-Hole Method	(9) Conv. Method	(10) Bunching-Hole Method	(11) Conv. Method
200K	1.0	2,105	0.531***	0.503***	17,000***	34,000***	0.291*	1.021**	0.333*	1.279**
			(0.036)	(0.031)	(4,260)	(7,037)	(0.109)	(0.430)	(0.186)	(0.636)
300K	1.5	8,108	0.860***	0.683***	27,500***	36,000**	0.107***	0.189	0.135***	0.258
			(0.029)	(0.020)	(4,492)	(13,766)	(0.039)	(0.188)	(0.050)	(0.266)
400K	2.5	11,111	0.712***	0.684***	20,500***	42,000***	0.065	0.171*	0.397***	0.184
			(0.037)	(0.033)	(6,760)	(11,451)	(0.044)	(0.095)	(0.046)	(0.131)
500K	2.5	14,286	0.512***	0.532***	30,500***	40,000**	0.062***	0.079	0.085***	0.111
			(0.031)	(0.020)	(9,960)	(9,376)	(0.016)	(0.053)	(0.017)	(0.074)
600K	2.5	17,647	0.841***	0.849***	31,500***	35,500**	0.025**	0.035	0.047***	0.060
			(0.035)	(0.031)	(5,742)	(10,952)	(0.019)	(0.030)	(0.015)	(0.047)

Notes: Considering the sample of self-employed individuals (non-rounders) from 2006 to 2009, the table presents estimates of friction share of individuals in dominated ranges who are unresponsive in columns (4)–(5), earnings responses to notches about friction in columns (6)–(7), and structural earnings elasticities based on either the parametric equation (8) in columns (8)–(9) or the reduced-form approximation (12) in columns (10)–(11). The bunching-hole method isolates observed bunching B by the inverse of the hole in the dominated range $315 - \alpha^a$ to estimate responses that are not attenuated by optimization frictions. The convergence method estimates responses based on the point of convergence between the observed and counterfactual densities. These two methods provide lower and upper bounds on the average structural elasticity in the population. Standard errors are shown in parentheses and stars indicate statistical significance level. * = 10% level, ** = 5% level, *** = 1% level.

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TABLE III STRUCTURAL EARNINGS ELASTICITIES FOR WAGE EARNERS (NON-ROUNDER SAMPLE)										
(1) Notch	(2) ATR Point	(3) Dominated Range Δx^b	Frictions α^a		Earnings Response Δx^a		Structural Elasticity ϵ		Reduced-Form Elasticity ϵ_R	
			(4) Full Range Δx^b	(5) Lower Range $\Delta x^{b/2}$	(6) Bunching-Hole Method	(7) Conv. Method	(8) Bunching-Hole Method	(9) Conv. Method	(10) Bunching-Hole Method	(11) Conv. Method
400K	1.0	4,145	0.914***	0.906***	9,000***	12,000	0.024***	0.031	0.024***	0.043
			(0.013)	(0.013)	(1,984)	(2,909)	(0.006)	(0.164)	(0.011)	(0.218)
500K	1.0	5,236	0.890***	0.899***	11,000***	16,500	0.033***	0.038	0.033***	0.052
			(0.008)	(0.008)	(2,045)	(15,760)	(0.010)	(0.168)	(0.009)	(0.300)
600K	1.0	9,574	0.813***	0.800***	24,000***	33,000**	0.034**	0.074	0.050***	0.084
			(0.011)	(0.011)	(5,310)	(16,361)	(0.019)	(0.113)	(0.023)	(0.133)
700K	1.5	11,351	0.870***	0.834***	12,500***	18,000	0.001	0.010	0.010**	0.020
			(0.010)	(0.010)	(1,928)	(17,476)	(0.013)	(0.064)	(0.003)	(0.070)

Notes: Considering the sample of wage earners (non-rounders) from 2006 to 2007, the table presents estimates of friction share of individuals in dominated ranges who are unresponsive in columns (4)–(5), earnings responses to notches about friction in columns (6)–(7), and structural earnings elasticities based on either the parametric equation (8) in columns (8)–(9) or the reduced-form approximation (12) in columns (10)–(11). The bunching-hole method isolates observed bunching B by the inverse of the hole in the dominated range $315 - \alpha^a$ to estimate responses that are not attenuated by optimization frictions. The convergence method estimates responses based on the point of convergence between the observed and counterfactual densities. These two methods provide lower and upper bounds on the average structural elasticity in the population. Standard errors are shown in parentheses and stars indicate statistical significance level. * = 10% level, ** = 5% level, *** = 1% level.

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(a) Table II: Structural earnings elasticities for self-employed individuals.

(b) Table III: Structural earnings elasticities for wage earners.

6.9 Figure VIII: Wage Earners, Non-Rounder Sample

Read:

- Wage earners bunch below notches, but the bunching is smaller than for self-employed individuals.
- Clear holes appear above several notches, consistent with the model's predicted missing mass.
- Estimated frictions are extremely large: about 90 percent of wage earners in dominated ranges remain unresponsive.
- The shape of the density is smoother than for the self-employed sample, consistent with less round-number reporting and less bracket-wide manipulation.

Economic message: Wage earners face stronger real or institutional constraints. Even large tax notches do not generate large immediate responses when income is set by wage contracts and third-party reporting.

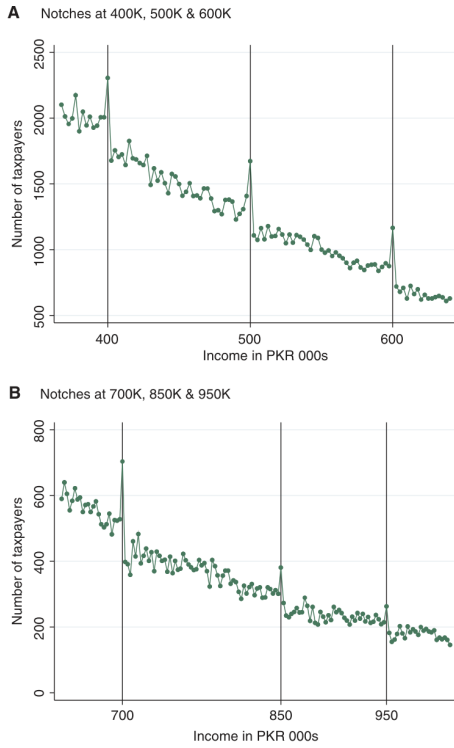


FIGURE VIII

Empirical and Counterfactual Distributions around Notches: Wage Earners
(Non-rounder Sample)

(a) Figure VIII, Panels A and B: Wage-earner distributions around notches.

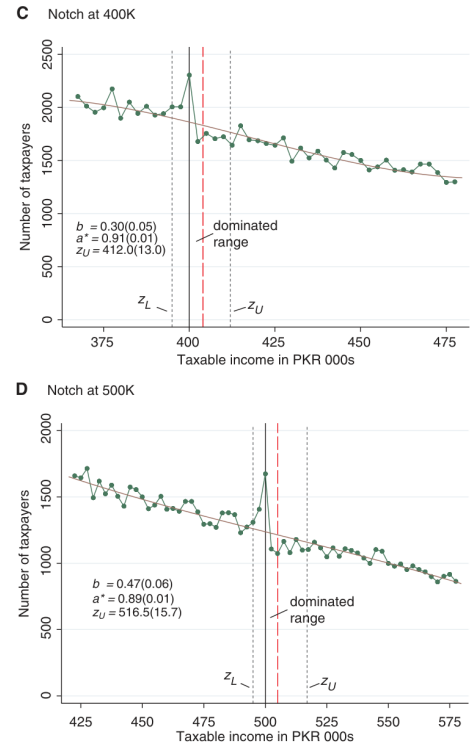


FIGURE VIII
Continued

long-dashed lines, and excluded ranges $[z_L, z_U]$ are marked by vertical short-dashed

(b) Figure VIII, Panels C and D: Wage-earner notches at 400K and 500K.

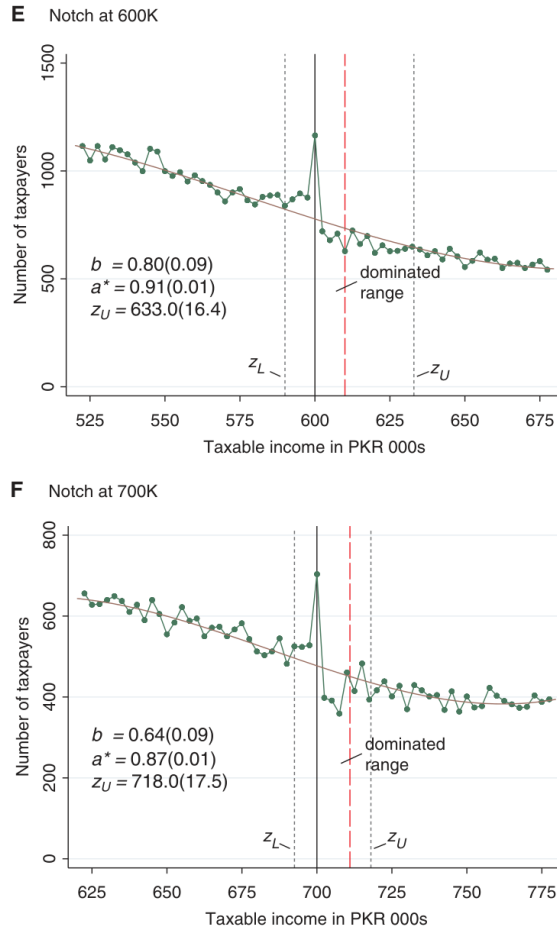


FIGURE VIII

Figure VIII, Panels E and F: Wage-earner notches at 600K and 700K.

6.10 Figure IX: Income-Composition Notch at 50 Percent Self-Employment Share

Read:

- The tax system classifies an individual as self-employed if self-employment income is at least 50 percent of total income.
- Because self-employed tax rates are higher, taxpayers have a strong incentive to keep the self-employment share below 50 percent.
- The figure shows large bunching below 50 percent and large missing mass above 50 percent.
- Bunching is diffuse rather than extremely sharp, suggesting that some taxpayers overshoot the cutoff substantially.
- The authors do not translate this response into a clean structural elasticity because the underlying adjustment may involve pure income shifting, total income changes, or both.

Economic message: Taxpayers also respond to notches in income composition, not just income levels. This supports the broader interpretation that tax notches distort reporting and tax-base composition.

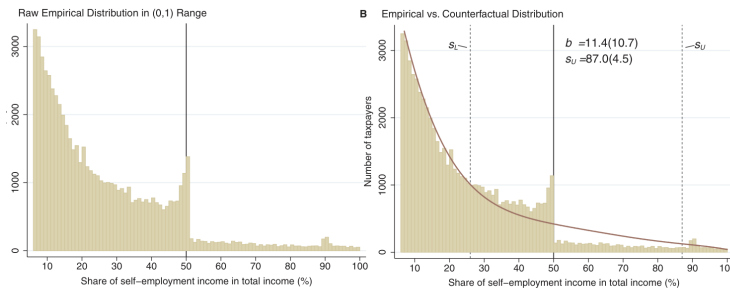


FIGURE IX
Empirical and Counterfactual Distributions of the Self-Employment Income Share: Behavioral Responses to the Income-Composition Notch at 50%

The figure shows the empirical distribution of the self-employment income share (in bars) and the counterfactual distribution (red line) for individuals with both self-employment income and wage income from 2006 to 2009. Panel A includes all observations with a self-employment income share between 0 and 1, and Panel B drops observations precisely at the cutoff value of 50%. The counterfactual is estimated by fitting a fifth-order polynomial to the empirical distribution, excluding data around the notch, as specified in equation (13). The notch point is marked by a vertical solid line and the excluded range $[s_L, s_U]$ is marked by vertical port-dashed lines. Bunching b is excess mass in the excluded range below the notch (in proportion to the average counterfactual frequency in this range), and the upper bound of the excluded range s_U has been estimated to ensure that missing mass equals bunching mass. Standard errors are shown in parentheses.

Figure IX: Empirical and counterfactual distributions of the self-employment income share.

7 Short summary

- The paper develops a theory and empirical method for using notches to identify optimization frictions and structural elasticities.
- A notch creates bunching below the cutoff and missing mass above it.
- The strictly dominated region above the cutoff should be empty in a frictionless world, so any observed mass there identifies frictions.
- The method combines bunching and missing mass to infer the frictionless response and the structural elasticity.
- The Pakistani income tax provides sharp notches because each bracket has a fixed average tax rate.
- Self-employed individuals bunch strongly below tax notches, but many remain in dominated regions.
- Wage earners bunch less and exhibit even larger frictions, consistent with third-party reporting and wage-contract constraints.
- Dynamics show that dominated behavior declines and bunching rises over time, suggesting learning and adjustment.
- Structural elasticities are generally modest even though frictionless notch responses are large.
- The main policy lesson is that notches can be highly inefficient because they create enormous implicit marginal tax rates around cutoffs.

8 Conclusion

8.1 Core conclusion

Kleven and Waseem show that notches provide a powerful way to identify both optimization frictions and structural elasticities. The central insight is that a notch creates a dominated region above the cutoff. In a frictionless model, this region must be empty. In the data, it is not empty. The remaining mass in dominated regions is direct evidence that many taxpayers do not fully optimize in response to strong tax incentives.

8.2 Economic intuition

The intuition is simple but powerful. A notch makes the budget set discontinuous. Just above the cutoff, earning slightly more can reduce after-tax income discretely. A taxpayer in this region could move down to the cutoff and obtain both more consumption and more leisure. Failure to do so reveals frictions. Once these frictions are measured, observed bunching can be inflated to recover the response that would occur absent frictions.

8.3 Incremental contribution revisited

The paper advances the public finance and development literature in four ways. First, it turns notches into a tool for measuring optimization frictions. Second, it separates observed and structural elasticities without imposing a fully parametric friction model. Third, it provides high-quality evidence from administrative data in Pakistan. Fourth, it demonstrates that notches can impose large efficiency costs even when estimated structural elasticities are small.

8.4 Implications for tax design

The results imply that policy makers should be cautious about using notches. A notch can create an implicit marginal tax rate far above 100 percent over a local income range. This generates strong bunching incentives, holes in the distribution, and inefficient income manipulation. Smoother tax schedules with kinks rather than notches can reduce these local distortions.

8.5 Implications for empirical work

The paper also warns against interpreting small observed elasticities as evidence of low structural responsiveness. Observed responses may be small because people face frictions, not because preferences are inelastic. Conversely, large bunching at a notch does not necessarily imply a large structural elasticity, because notches create extremely strong incentives.

8.6 Limitations

The method focuses on local intensive-margin responses among tax filers. It does not fully capture extensive-margin behavior such as entry into informality, exit from filing, or other broad tax-base responses. The data cover registered filers, who may be more compliant or better monitored than

the broader population. Some responses may also involve reporting manipulation rather than real labor supply or productivity changes.

8.7 Takeaway

The enduring takeaway is that notches reveal both behavior and nonbehavior. Bunching shows that some taxpayers respond strongly to incentives; dominated-region mass shows that many taxpayers do not. The combination allows the researcher to recover structural elasticities while acknowledging the frictions that shape observed behavior.